

Part 6 – Assignment

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This assignment is worth 50 marks. Solutions must be written in **R** markdown and submitted *both* as Rmd-file and compiled pdf-file. Feel free to copy as much code from the lecture material as you like if it helps solve a problem. Use embedded **R**-code for numerical calculations.

These libraries should be sufficient, but you are welcome to add others:

```
library(magicaxis)
library(cubature)
library(cooltools)
library(plotrix)
library(hyper.fit)
library(LaplacesDemon)
```

Hidden triangles [10 marks]

Consider a 3D Euclidean space filled with many randomly positioned and randomly oriented Pythagorean triangles with side-lengths $3s$, $4s$ and $5s$. The length scale s is unknown but identical for all triangles. Even though there are infinitely many such random triangles in the infinite space, you only have access to the positions of the corners that happen to lie within a radius $R \in [0.9, 1.0]$ from the origin of the coordinate system. These positions are stored in the file `data/hiddentriangles.csv`.

- Plot the 2PCF $\xi(r)$ with Poisson errors on a grid from $r = 0$ to $r = 0.4$, with a step $\Delta r = 0.005$, using the Landy-Szalay estimator with 10^4 random reference points in the shell of radius $R \in [0.9, 1.0]$. [7 marks]
- What's your best guess for the value of s ? [3 marks]

Matter power spectrum [20 marks]

The file `data/L210N1536.bin` is a binary file encoding an array with N -by- N -by- N elements, where $N = 128$. This array is a discretised representation of the density contrast $\delta(\mathbf{x})$ at redshift $z = 0$, of a periodic simulation cube of side length $L = 210\text{Mpc}/h$, simulated with GADGET using 1536^3 particles (Elahi et al. 2018, MNRAS 475). The data can be loaded into **R** using the function `loadbin(filename, c(128, 128, 128), signed=TRUE)` of the **cooltools** package.

- Visualise the density field, by plotting the first layer in the third dimension (`[,1]`) of $f = (1 + \delta)^{0.2}$ using a linear gray scale from full black to full white. Make sure that the image is a square by setting `par(pty='s')`. [5 marks]
- Compute the 2PCF $\xi(r)$ via the inverse FT of the 3D power spectrum $P(\mathbf{k}) = |\hat{\delta}(\mathbf{k})|^2$. Show $\xi(r)$ in a log-log plot, plotting only the points $r < 20\text{Mpc}/h$ that are multiples of the grid spacing $\Delta r = L/N$. Hint: Use equation (5) of part 6, section 2, and pay attention to the fact that the sum over $\mathbf{k}$ must go over all N^3 cells in Fourier space. [10 marks]
- Use `hyper.fit` to fit a straight line in log-log space through the points of the previous problem (neglect errors). Plot the line on top of the points. What is slope of the line (i.e. the power law index in linear space) and the $1\text{-}\sigma$ uncertainty of the slope? Your solution must be correct within 0.001. [5 marks]

A new field [20 marks]

It is the year 2034 and research in cosmology has taken an exciting turn: an exotic background radiation has been inferred from anomalies in Square Kilometre Array data. This radiation is made of “quosons”, a novel elementary particle, now detectable directly via the international quoson detector. This detector is equally sensitive to quosons coming from any direction. During its first observing run, the detector has registered a few hundred quosons. Their (θ, φ) -coordinates are stored in the file `data/quosons.csv`.

The quoson radiation is thought to be isotropic in the CMB rest-frame. However, the detector moves relative to this rest-frame, making it more likely to detect quosons in the direction the detector is heading to. The probability density *per unit solid angle* of detecting a quoson in any direction (θ, φ) can be modelled as

$$\rho(\theta, \varphi) = \frac{1}{4\pi} + a Y_{D(\theta_0, \varphi_0)}(\theta, \varphi),$$

where a is a numerical constant and $Y_{D(\theta_0, \varphi_0)}$ is the dipole spherical harmonic, whose mode lies at $(\theta, \varphi) = (\theta_0, \varphi_0)$. The constant a depends on unknown quoson properties, but has been constrained to the interval $a \in [0.05, 0.15]$ in laboratory experiments.

- a) Plot the quoson positions in a Mollweide projection with $(\varphi, \theta) = (0, \pi/2)$ at the centre. [5 marks]
- b) Compute the MAP solution and individual 1-sigma uncertainties for the three parameters $a \in [0.05, 0.15]$, $\theta_0 \in [0, \pi)$ and $\varphi_0 \in [0, 2\pi)$. Assume a uniform prior for $a \in [0.05, 0.15]$ and an isotropic prior for the dipole direction. All values should be correct within an absolute error of 0.001. Hints: Carefully reflect what an *isotropic* prior means for the polar coordinate θ . Use `optim` with the method “L-BFGS-B” to specify the parameter ranges. [10 marks]
- c) Sample the posterior $\rho(a, \theta_0, \varphi_0 | \text{data})$ using MCMC. To do so call `set.seed(1)` and use Laplace’s Demon with initial values $(0.1, \pi/2, \pi)$, the “CHARM” algorithm, 10^4 iterations and a thinning parameter of 10. Display the posterior sample using `magtri`. [5 marks]